Chemistry V2 — Core Study Guide

Source-grounded cold-start learner candidate. Main teaching is followed by Appendix A Core Practice, Appendix B Core Solutions and Appendix C Printable Handout.

Attach a role only after tracking the species

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Treat the equation as a before/after map of the same chemical entities.

Explain the idea

A role is proved from what a species does in this equation, not from reagent reputation.

Representation path

- Track changing species first.
- Separate unchanged or spectator information.
- Attach a role only after the change is established.
- Use the symbolic representation.
- Read the formula notation explicitly.
- Read the ionic charge explicitly.

Track each element from before to after

$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

BEFORE		AFTER
Zn	→	Zn^{2+}
Cu^{2+}		Cu

---- tracked element changes host species ---- tracked element unchanged

Elements tracked: Cu, Zn

Sort the species before naming any role

$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

Reactant species	Zn Cu^{2+}
Species that changes	fill only from the evidence above
Spectator (if present)	fill only from the evidence above
Product species	Zn^{2+} Cu

Attach a role only after identifying the changing reactant species.

Things to know

Equation/source evidence is authoritative; memorized reagent reputation is not.

Reconstruct the reasoning

- Identify reactant species.
- Track each through the equation.
- Mark change/no change.
- Attach the role to the reactant evidence.
- Verify spectators are not mislabelled.

Worked example

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Work on this practice instance before attempting the transfer questions.

- Track the reactant species first.
- Use its recorded change/effect as evidence.

- Attach the requested role and exclude spectators.

Check

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: A familiar reagent has one permanent role.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: A minimal pair differing on exactly one decisive feature.

- Elicit the wrong model.
- Show the discriminating contrast.
- State the repaired model.
- Retry immediately.

Retry: After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
Zn		Zn
Predict this case before reading on.		Predict this case before reading on.

Try with me

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Use the first-move and representation cues shown here.

Faded practice

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. One cue has been removed; choose the next move yourself.

Check your knowledge

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Try it independently using only the information in the task.

Verification

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Check that the overall charge is preserved or interpreted correctly.
- [ ] Check that the same chemical species is being tracked.
- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: same role logic in a new equation. Attempt the transfer questions only after this learning step.

Check atom conservation explicitly

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Frame conservation as an accounting check on a represented process.

Explain the idea

Count the exact thing the route says must be conserved; different ledgers answer different questions.

Representation path

- Create before/after ledgers for the required invariant.
- Keep atom count, charge and species identity as separate ledgers.
- Use the symbolic representation.
- Read the formula notation explicitly.

Atom ledger before and after

$2H_2 + O_2 \rightarrow 2H_2O$

Element	Before	After	Matched?
H	4 ▮▮▮▮	4 ▮▮▮▮	✓ same
O	2 ▮▮	2 ▮▮	✓ same

Fill or compare the required ledger and locate the first mismatch.

Things to know

Choose the conservation check required by the question.

Reconstruct the reasoning

- Name the ledger.
- Count before/after.
- Compare.
- Locate the first mismatch.
- Recheck after repair.

Worked example

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Work on this practice instance before attempting the transfer questions.

- Choose the atom-count ledger.
- Count each required atom before and after.
- Compare counts and identify any mismatch.

Check

- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: A balanced-looking expression has passed every chemical check.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: Contrast an atom-count mismatch with a charge or species-identity issue.

- Separate the ledgers.
- Repair the failed ledger only.
- Run the remaining required checks.

Retry: Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
$2H_2$		$2H_2$
Predict this case before reading on.		Predict this case before reading on.

Try with me

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Use the first-move and representation cues shown here.

Faded practice

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. One cue has been removed; choose the next move yourself.

Check your knowledge

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Try it independently using only the information in the task.

Verification

- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Count each required atom on both sides.
- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: new process with the same conservation obligation. Attempt the transfer questions only after this learning step.

Check the rule, condition and exception

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Show the default rule and the gate that can block or modify it.

Explain the idea

A remembered rule is incomplete until the stated condition or exception is checked.

Representation path

- Rule cue first.
- Condition/exception gate second.
- Decision only after the gate check.
- Use the symbolic representation.

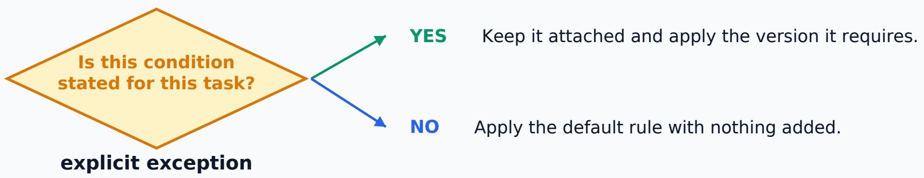
Order in which the rule is consulted

Rule selection before condition/exception decision.

1. Read the rule as stated for this task
2. Check the stated condition: explicit exception
3. Check the stated exception: explicit exception supplied by source
4. Apply the rule, or apply the override the condition requires

Stop at the first step whose stated condition decides the case.

Condition gate before the rule applies



Things to know

Only upstream-recorded rules, conditions and exceptions may populate the gate.

Reconstruct the reasoning

- Select the rule.
- Read the condition/exception.
- Test applicability.
- Apply or withhold the rule.
- Verify the condition remains attached.

Worked example

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Work on this practice instance before attempting the transfer questions.

- Select the recorded rule.
- Check the condition/exception gate.
- Apply or withhold the rule accordingly.

Check

- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: Remembering the default rule is enough.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: A minimal pair differing on exactly one decisive feature.

- Elicit the wrong model.
- Show the discriminating contrast.
- State the repaired model.
- Retry immediately.

Retry: Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
Predict this case before reading on.		Predict this case before reading on.

Try with me

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Use the first-move and representation cues shown here.

Faded practice

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. One cue has been removed; choose the next move yourself.

Check your knowledge

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Try it independently using only the information in the task.

Verification

- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Check that every stated condition or exception is still attached.
- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: same rule with a less obvious condition/exception. Attempt the transfer questions only after this learning step.

Classify a change from the stated evidence

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Anchor classification in the supplied evidence or process description.

Explain the idea

Name the evidence before naming the class.

Representation path

- Evidence features lead to a criterion; the criterion leads to the class.
- Use the stated macroscopic evidence.
- Use the symbolic representation.
- Keep the recorded observation separate from interpretation.
- Keep physical-state information attached.

What was actually observed

RECORDED AT THE MACROSCOPIC LEVEL

Expose the source-recorded observable evidence before a chemical claim or classification.

Nothing has been inferred yet — this box stays empty until you decide.

Evidence, link, claim

EVIDENCE

Separate observation, claim and reasoning link so overclaiming is inspectable.

→

LINK

Which evidence supports which part of the claim.

→

CLAIM

Read observation first, then test each claim against the evidence link.

An inference is only as strong as the recorded evidence behind it.

Things to know

Classify only from the stated evidence and the rule taught here.

Reconstruct the reasoning

- Extract evidence.
- Select the criterion.
- Compare.
- Classify.
- Verify that the evidence supports the label.

Worked example

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Work on this practice instance before attempting the transfer questions.

- Extract the supplied evidence.
- Apply the authorized criterion.
- Classify only after the evidence matches the criterion.

Check

- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: A familiar word in the prompt is enough to classify the process.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: Use two cases with similar wording but different decisive evidence.

- Return to the evidence.
- Apply the criterion explicitly.
- Retry on the near contrast.

Retry: State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
Predict this case before reading on.		Predict this case before reading on.

Try with me

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Use the first-move and representation cues shown here.

Faded practice

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. One cue has been removed; choose the next move yourself.

Check your knowledge

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Try it independently using only the information in the task.

Verification

- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: near-neighbour classification with new evidence. Attempt the transfer questions only after this learning step.

Separate ionic charge from subscripts

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Use a side-by-side notation pair rather than a warning alone.

Explain the idea

The position of the numeral is evidence for what it means.

Representation path

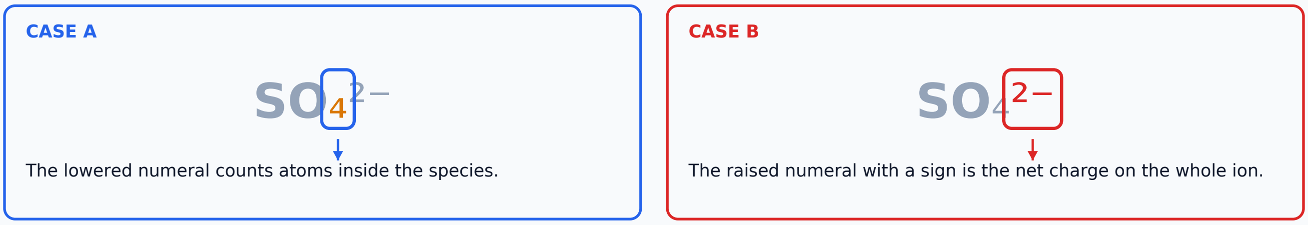
- Hold element symbols fixed.
- Change one numeral position only.
- Use the symbolic representation.
- Read the formula notation explicitly.
- Read the ionic charge explicitly.

Anatomy of the formula

Notation position and its local chemical meaning.

Charge position versus subscript position

Same species, same symbols — only the position you read changes.



Compare two cases and state what the numeral means in each position.

Things to know

Read coefficient, subscript and charge as different notation roles.

Reconstruct the reasoning

- Present a minimal pair.
- Identify the only changed feature.
- Explain the changed meaning.
- Retry without labels.

Worked example

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Work on this practice instance before attempting the transfer questions.

- Locate the superscript charge separately from subscripts/coefficients.
- Read the ionic charge.
- Verify the species identity is unchanged.

Check

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: The numeral itself carries a fixed meaning wherever it appears.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: A minimal pair differing on exactly one decisive feature.

- Elicit the wrong model.
- Show the discriminating contrast.
- State the repaired model.
- Retry immediately.

Retry: Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
Fe^{3+}		Fe^{3+}
Predict this case before reading on.		Predict this case before reading on.

Try with me

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Use the first-move and representation cues shown here.

Faded practice

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. One cue has been removed; choose the next move yourself.

Check your knowledge

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Try it independently using only the information in the task.

Verification

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

Run these checks before accepting

- [ ] Check that the overall charge is preserved or interpreted correctly.
- [ ] Check that the same chemical species is being tracked.

Transfer family: notation-position discrimination under reduced cues. Attempt the transfer questions only after this learning step.

Keep reaction conditions attached

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Show the default rule and the gate that can block or modify it.

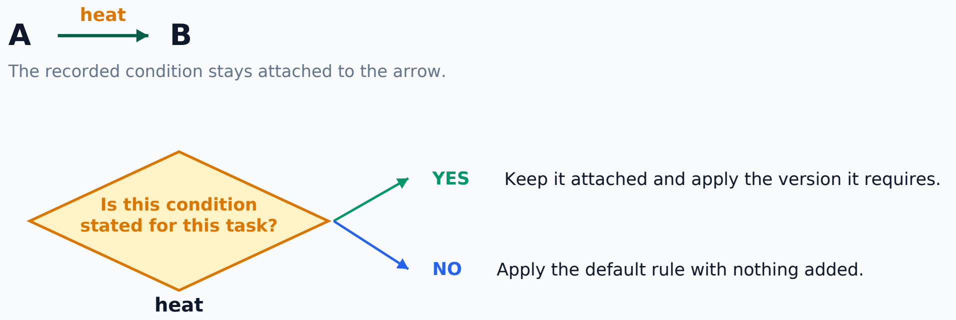
Explain the idea

A remembered rule is incomplete until the stated condition or exception is checked.

Representation path

- Rule cue first.
- Condition/exception gate second.
- Decision only after the gate check.
- Use the symbolic representation.
- Use the condition representation.
- Read the formula notation explicitly.

Condition gate before the rule applies



Things to know

Only upstream-recorded rules, conditions and exceptions may populate the gate.

Reconstruct the reasoning

- Select the rule.
- Read the condition/exception.
- Test applicability.
- Apply or withhold the rule.
- Verify the condition remains attached.

Worked example

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Work on this practice instance before attempting the transfer questions.

- Locate the recorded condition.
- Keep it attached to the process representation.
- Check that the rewritten/read representation preserves it.

Check

- Check that every stated condition or exception is still attached.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: Remembering the default rule is enough.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: Two cases use the same rule but differ only in the decisive condition/exception.

- Re-run the gate.
- Name the decisive condition.
- Retry on a close contrast.

Retry: Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
A		A
Predict this case before reading on.		Predict this case before reading on.

Try with me

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Use the first-move and representation cues shown here.

Faded practice

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. One cue has been removed; choose the next move yourself.

Check your knowledge

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Try it independently using only the information in the task.

Verification

- Check that every stated condition or exception is still attached.

Run these checks before accepting

- [ ] Check that every stated condition or exception is still attached.

Transfer family: same rule with a less obvious condition/exception. Attempt the transfer questions only after this learning step.

Connect apparatus, property and method

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Start from the property/evidence needed, then connect it to the apparatus or method feature.

Explain the idea

Choose the method because of the property it exploits, not because the diagram looks familiar.

Representation path

- Target property → apparatus feature → expected observation/use.
- Use the stated macroscopic evidence.
- Read the figure as part of the evidence.

Property, apparatus, intended observation

PROPERTY USED
Property → apparatus feature → intended observation or use.

↓

APPARATUS / METHOD
Connect apparatus or method choice to the source-authorized property or observation it uses.

↓

INTENDED OBSERVATION
Trace the method choice back to the property/evidence it is meant to exploit.

Things to know

Only apparatus/method properties authorized by source may be used.

Reconstruct the reasoning

- Identify target property.
- Identify apparatus feature.
- Link feature to property.
- Verify against the goal/observation.

Worked example

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Work on this practice instance before attempting the transfer questions.

- Identify the target property or observation.
- Identify the apparatus/method feature.
- Explain the property-feature link.

Check

- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: Recognizing the apparatus shape is enough.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: Compare two visually similar method representations that differ on the decisive property link.

- Name the property first.
- Trace it to the apparatus feature.
- Retry with labels removed.

Retry: Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
Predict this case before reading on.		Predict this case before reading on.

Try with me

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Use the first-move and representation cues shown here.

Faded practice

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. One cue has been removed; choose the next move yourself.

Check your knowledge

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Try it independently using only the information in the task.

Verification

- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: same property-method link in a different representation. Attempt the transfer questions only after this learning step.

Read a chemical formula before calculating

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Treat the formula as a structured message, not a string of characters.

Explain the idea

Read position first, then meaning: a lower numeral, a leading numeral and an upper charge do different jobs.

Representation path

- Point separately to symbol, subscript, coefficient and charge positions.
- Map each position to its local meaning.
- Use the symbolic representation.
- Read the formula notation explicitly.
- Read the ionic charge explicitly.

Things to know

Do not infer chemical facts beyond the notation and source-authorized rules.

Reconstruct the reasoning

- Locate notation features.
- Name each feature.
- State its local meaning.
- Check the whole species identity.

Worked example

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Work on this practice instance before attempting the transfer questions.

- Locate each notation feature by position.
- State the local meaning of each feature.
- Check the complete species notation.

Check

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: All nearby numerals mean the same thing.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: A minimal pair differing on exactly one decisive feature.

- Elicit the wrong model.
- Show the discriminating contrast.
- State the repaired model.
- Retry immediately.

Retry: Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
Predict this case before reading on.		Predict this case before reading on.

Try with me

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Use the first-move and representation cues shown here.

Faded practice

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. One cue has been removed; choose the next move yourself.

Check your knowledge

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Try it independently using only the information in the task.

Verification

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

Run these checks before accepting

- [ ] Check that the overall charge is preserved or interpreted correctly.
- [ ] Check that the same chemical species is being tracked.

Transfer family: formula/charge discrimination in a new species. Attempt the transfer questions only after this learning step.

Separate observation from inference

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Start with what was actually observed before naming a chemical conclusion.

Explain the idea

An observation is evidence; a claim must not say more than that evidence supports.

Representation path

- Record observation.
- State claim separately.
- Use an explicit evidence-to-claim link.
- Use the stated macroscopic evidence.
- Keep the recorded observation separate from interpretation.

What was actually observed

RECORDED AT THE MACROSCOPIC LEVEL

Expose the source-recorded observable evidence before a chemical claim or classification.

Nothing has been inferred yet — this box stays empty until you decide.

Observation kept apart from inference

Literal evidence versus interpreted chemical claim.

OBSERVATION — what was recorded	INFERENCE — what is claimed from it
Hold observation and inference in separate cells so one cannot be silently substituted for the other.	

Place each statement in the correct column and justify the inference link.

Things to know

Source observations and authorized conclusions constrain the reasoning chain.

Reconstruct the reasoning

- Record the observation literally.
- State a candidate claim.
- Explain the link.
- Test for overclaim.

Worked example

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Work on this practice instance before attempting the transfer questions.

- Record the observation literally.
- State the claim separately.
- Explain exactly how the observation supports the claim.

Check

- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: A chemically plausible claim is automatically supported.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: A minimal pair differing on exactly one decisive feature.

- Elicit the wrong model.
- Show the discriminating contrast.
- State the repaired model.
- Retry immediately.

Retry: Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

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CASE A	≠	CASE B
Predict this case before reading on.		Predict this case before reading on.

Try with me

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Use the first-move and representation cues shown here.

Faded practice

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. One cue has been removed; choose the next move yourself.

Check your knowledge

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Try it independently using only the information in the task.

Verification

- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: new observation requiring cautious inference. Attempt the transfer questions only after this learning step.

Track the same reacting species

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

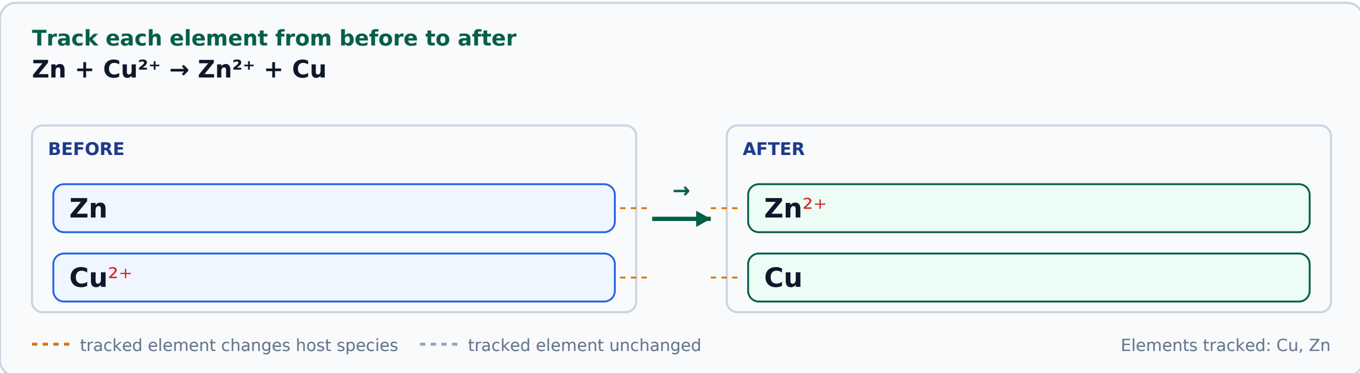
Treat the equation as a before/after map of the same chemical entities.

Explain the idea

A role is proved from what a species does in this equation, not from reagent reputation.

Representation path

- Track changing species first.
- Separate unchanged or spectator information.
- Attach a role only after the change is established.
- Use the symbolic representation.
- Read the formula notation explicitly.
- Read the ionic charge explicitly.



Things to know

Equation/source evidence is authoritative; memorized reagent reputation is not.

Reconstruct the reasoning

- Identify reactant species.
- Track each through the equation.
- Mark change/no change.
- Attach the role to the reactant evidence.
- Verify spectators are not mislabelled.

Worked example

Track each supplied reacting species from before to after the process before assigning any role label. Work on this practice instance before attempting the transfer questions.

- Identify reactant species.
- Track each corresponding species after the process.
- Mark changed and unchanged species.

Check

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: A familiar reagent has one permanent role.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: Compare a changed species with an unchanged/spectator species in the same representation.

- Return to species tracking.
- Name the self-change before the role.
- Retry with a new equation/representation.

Retry: Track each supplied reacting species from before to after the process before assigning any role label. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
Zn		Zn
Predict this case before reading on.		Predict this case before reading on.

Try with me

Track each supplied reacting species from before to after the process before assigning any role label. Use the first-move and representation cues shown here.

Faded practice

Track each supplied reacting species from before to after the process before assigning any role label. One cue has been removed; choose the next move yourself.

Check your knowledge

Track each supplied reacting species from before to after the process before assigning any role label. Try it independently using only the information in the task.

Verification

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Check that the overall charge is preserved or interpreted correctly.
- [ ] Check that the same chemical species is being tracked.
- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: same role logic in a new equation. Attempt the transfer questions only after this learning step.

Translate between particles and symbols

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Begin with the representation the learner can directly inspect before introducing another representation.

Explain the idea

Symbols are a compressed way to describe the same chemical entities; translation must not change what is present.

Representation path

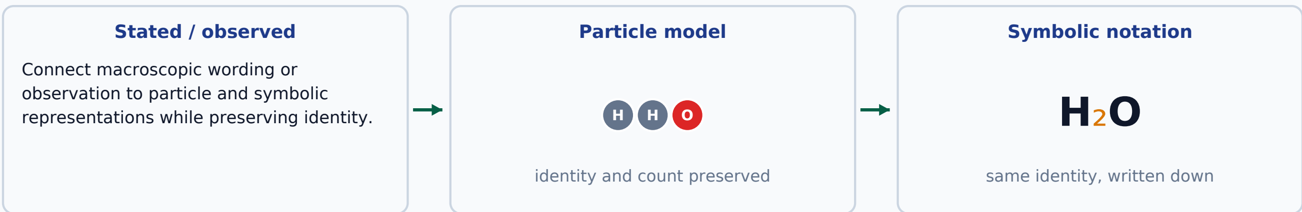
- Name the current representation level.
- Hold chemical identity fixed while translating.
- Return to the start representation to check equivalence.
- Use the particulate representation.
- Use the symbolic representation.
- Read the figure as part of the evidence.
- Read the formula notation explicitly.

Particle view of the same substance

School-level particle-count model — composition and count only.

Same chemical entity across three views

Which chemical entity remains the same across representation levels.



Trace the same entity through macro, particle and symbolic panels.

Things to know

Preserve identity, count and any recorded condition while changing representation.

Reconstruct the reasoning

- Identify the source representation.
- Mark invariant entities.
- Translate one feature at a time.
- Cross-check against the starting representation.

Worked example

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Work on this practice instance before attempting the transfer questions.

- Identify the starting representation level.
- Mark invariant entities/counts.
- Translate one feature at a time.
- Cross-check against the starting representation.

Check

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: Changing notation is harmless if the picture looks similar.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: Compare a faithful translation with one that changes a subscript, charge or species identity.

- Return to the source representation.
- Repair the first changed feature.
- Retry in the reverse direction.

Retry: Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
H_2O		H_2O
Predict this case before reading on.		Predict this case before reading on.

Try with me

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Use the first-move and representation cues shown here.

Faded practice

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. One cue has been removed; choose the next move yourself.

Check your knowledge

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Try it independently using only the information in the task.

Verification

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Check that the same chemical species is being tracked.
- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: same chemistry in a different representation. Attempt the transfer questions only after this learning step.

Verify a chemical representation before accepting it

Orient / activate

Identify the target and state the first chemically meaningful move: Identify the target.

Make checking a separate pass, not a repetition of the final answer.

Explain the idea

A tidy final answer is not chemically verified until an independent check is applied.

Representation path

- Run only applicable checks: species identity, atoms, charge, units, state/condition and plausibility.
- Use the symbolic representation.

Run these checks before accepting

- [ ] Check that the same chemical species is being tracked.
- [ ] Check that the final conclusion follows from the evidence and rule used.

Things to know

Verification obligations are inherited from source/problem-family authority.

Reconstruct the reasoning

- Name the required checks.
- Run each independently.
- Locate any failure.
- Repair.
- Re-run.

Worked example

Run the required independent chemical checks on the supplied result or representation before accepting it. Work on this practice instance before attempting the transfer questions.

- Name the required verification checks.
- Run each check independently.
- Repair any failed check and re-run it.

Check

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Concept helper

A first move should expose the governing evidence, representation or rule without giving away the result.

Misconception clinic

Wrong model: Getting an answer means it has been checked.

The shortcut can look sufficient because it uses a familiar surface cue before the decisive chemical evidence is checked.

Contrast: Contrast a neat-looking result with one that passes the required chemical invariant.

- Choose one independent check.
- Apply it.
- Repair and repeat.

Retry: Run the required independent chemical checks on the supplied result or representation before accepting it. Retry after applying the repaired model to a close but newly authored instance.

Two cases, one decisive difference

The one feature that changes the correct interpretation or decision.

HELD CONSTANT: Hold context constant while changing one decisive chemical feature to expose a wrong model.

CASE A	≠	CASE B
Predict this case before reading on.		Predict this case before reading on.

Try with me

Run the required independent chemical checks on the supplied result or representation before accepting it. Use the first-move and representation cues shown here.

Faded practice

Run the required independent chemical checks on the supplied result or representation before accepting it. One cue has been removed; choose the next move yourself.

Check your knowledge

Run the required independent chemical checks on the supplied result or representation before accepting it. Try it independently using only the information in the task.

Verification

- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

Run these checks before accepting

- [ ] Check that the same chemical species is being tracked.
- [ ] Check that the final conclusion follows from the evidence and rule used.

Transfer family: verification on a new representation/problem family. Attempt the transfer questions only after this learning step.

Appendix A — Core Practice

Practice 1 — Attach a role only after tracking the species

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Use the first-move and representation cues shown here.

Workspace: ____

Practice 2 — Attach a role only after tracking the species

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 3 — Attach a role only after tracking the species

After tracking the supplied species, attach the requested role only to the reactant species whose equation evidence supports it. Try it independently using only the information in the task.

Workspace: ____

Practice 4 — Check atom conservation explicitly

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Use the first-move and representation cues shown here.

Workspace: ____

Practice 5 — Check atom conservation explicitly

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 6 — Check atom conservation explicitly

Use an atom-count ledger to check the supplied process representation. State the first mismatch or confirm the required conservation check. Try it independently using only the information in the task.

Workspace: ____

Practice 7 — Check the rule, condition and exception

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Use the first-move and representation cues shown here.

Workspace: ____

Practice 8 — Check the rule, condition and exception

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 9 — Check the rule, condition and exception

Apply the supplied rule only after checking the recorded condition or exception. State which feature controls the decision. Try it independently using only the information in the task.

Workspace: ____

Practice 10 — Classify a change from the stated evidence

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Use the first-move and representation cues shown here.

Workspace: ____

Practice 11 — Classify a change from the stated evidence

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 12 — Classify a change from the stated evidence

State the decisive supplied evidence first, then classify the process only from that evidence and the authorized criterion. Try it independently using only the information in the task.

Workspace: ____

Practice 13 — Separate ionic charge from subscripts

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Use the first-move and representation cues shown here.

Workspace: ____

Practice 14 — Separate ionic charge from subscripts

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 15 — Separate ionic charge from subscripts

Read the supplied ion notation. Distinguish ionic charge from subscripts or coefficients and state the charge without changing the species. Try it independently using only the information in the task.

Workspace: ____

Practice 16 — Keep reaction conditions attached

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Use the first-move and representation cues shown here.

Workspace: ____

Practice 17 — Keep reaction conditions attached

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 18 — Keep reaction conditions attached

Read or rewrite the supplied reaction while preserving the recorded condition and explain why the condition cannot be dropped from the representation. Try it independently using only the information in the task.

Workspace: ____

Practice 19 — Connect apparatus, property and method

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Use the first-move and representation cues shown here.

Workspace: ____

Practice 20 — Connect apparatus, property and method

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 21 — Connect apparatus, property and method

Use the supplied apparatus representation to link the target property or observation to the relevant method feature. Try it independently using only the information in the task.

Workspace: ____

Practice 22 — Read a chemical formula before calculating

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Use the first-move and representation cues shown here.

Workspace: ____

Practice 23 — Read a chemical formula before calculating

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 24 — Read a chemical formula before calculating

Read the supplied formula as structured notation. Identify symbols, any coefficient, any subscript and any charge before interpreting it. Try it independently using only the information in the task.

Workspace: ____

Practice 25 — Separate observation from inference

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Use the first-move and representation cues shown here.

Workspace: ____

Practice 26 — Separate observation from inference

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 27 — Separate observation from inference

Write the supplied observation and the supported inference separately, then state the evidence link without overclaiming. Try it independently using only the information in the task.

Workspace: ____

Practice 28 — Track the same reacting species

Track each supplied reacting species from before to after the process before assigning any role label. Use the first-move and representation cues shown here.

Workspace: ____

Practice 29 — Track the same reacting species

Track each supplied reacting species from before to after the process before assigning any role label. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 30 — Track the same reacting species

Track each supplied reacting species from before to after the process before assigning any role label. Try it independently using only the information in the task.

Workspace: ____

Practice 31 — Translate between particles and symbols

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Use the first-move and representation cues shown here.

Workspace: ____

Practice 32 — Translate between particles and symbols

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 33 — Translate between particles and symbols

Translate the supplied particle or symbolic representation into the requested representation while preserving chemical identity and count. Try it independently using only the information in the task.

Workspace: ____

Practice 34 — Verify a chemical representation before accepting it

Run the required independent chemical checks on the supplied result or representation before accepting it. Use the first-move and representation cues shown here.

Workspace: ____

Practice 35 — Verify a chemical representation before accepting it

Run the required independent chemical checks on the supplied result or representation before accepting it. One cue has been removed; choose the next move yourself.

Workspace: ____

Practice 36 — Verify a chemical representation before accepting it

Run the required independent chemical checks on the supplied result or representation before accepting it. Try it independently using only the information in the task.

Workspace: ____

Appendix B — Core Solutions

Solution 1 — Attach a role only after tracking the species

- Track the reactant species first.
- Use its recorded change/effect as evidence.
- Attach the requested role and exclude spectators.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 2 — Attach a role only after tracking the species

- Track the reactant species first.
- Use its recorded change/effect as evidence.
- Attach the requested role and exclude spectators.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 3 — Attach a role only after tracking the species

- Track the reactant species first.
- Use its recorded change/effect as evidence.
- Attach the requested role and exclude spectators.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 4 — Check atom conservation explicitly

- Choose the atom-count ledger.
- Count each required atom before and after.
- Compare counts and identify any mismatch.
- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 5 — Check atom conservation explicitly

- Choose the atom-count ledger.
- Count each required atom before and after.
- Compare counts and identify any mismatch.
- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 6 — Check atom conservation explicitly

- Choose the atom-count ledger.
- Count each required atom before and after.
- Compare counts and identify any mismatch.
- Count each required atom on both sides.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 7 — Check the rule, condition and exception

- Select the recorded rule.
- Check the condition/exception gate.
- Apply or withhold the rule accordingly.
- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve the explicit exception supplied with the task.

Solution 8 — Check the rule, condition and exception

- Select the recorded rule.
- Check the condition/exception gate.
- Apply or withhold the rule accordingly.
- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve the explicit exception supplied with the task.

Solution 9 — Check the rule, condition and exception

- Select the recorded rule.
- Check the condition/exception gate.
- Apply or withhold the rule accordingly.
- Check that every stated condition or exception is still attached.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve the explicit exception supplied with the task.

Solution 10 — Classify a change from the stated evidence

- Extract the supplied evidence.
- Apply the authorized criterion.
- Classify only after the evidence matches the criterion.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 11 — Classify a change from the stated evidence

- Extract the supplied evidence.

- Apply the authorized criterion.
- Classify only after the evidence matches the criterion.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 12 — Classify a change from the stated evidence

- Extract the supplied evidence.
- Apply the authorized criterion.
- Classify only after the evidence matches the criterion.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 13 — Separate ionic charge from subscripts

- Locate the superscript charge separately from subscripts/coefficients.
- Read the ionic charge.
- Verify the species identity is unchanged.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 14 — Separate ionic charge from subscripts

- Locate the superscript charge separately from subscripts/coefficients.
- Read the ionic charge.
- Verify the species identity is unchanged.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 15 — Separate ionic charge from subscripts

- Locate the superscript charge separately from subscripts/coefficients.
- Read the ionic charge.
- Verify the species identity is unchanged.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 16 — Keep reaction conditions attached

- Locate the recorded condition.
- Keep it attached to the process representation.
- Check that the rewritten/read representation preserves it.
- Check that every stated condition or exception is still attached.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve and apply: heat

Solution 17 — Keep reaction conditions attached

- Locate the recorded condition.
- Keep it attached to the process representation.
- Check that the rewritten/read representation preserves it.
- Check that every stated condition or exception is still attached.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve and apply: heat

Solution 18 — Keep reaction conditions attached

- Locate the recorded condition.
- Keep it attached to the process representation.
- Check that the rewritten/read representation preserves it.
- Check that every stated condition or exception is still attached.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

Preserve and apply: heat

Solution 19 — Connect apparatus, property and method

- Identify the target property or observation.
- Identify the apparatus/method feature.
- Explain the property-feature link.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 20 — Connect apparatus, property and method

- Identify the target property or observation.
- Identify the apparatus/method feature.
- Explain the property-feature link.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 21 — Connect apparatus, property and method

- Identify the target property or observation.
- Identify the apparatus/method feature.
- Explain the property-feature link.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 22 — Read a chemical formula before calculating

- Locate each notation feature by position.
- State the local meaning of each feature.
- Check the complete species notation.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 23 — Read a chemical formula before calculating

- Locate each notation feature by position.
- State the local meaning of each feature.
- Check the complete species notation.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 24 — Read a chemical formula before calculating

- Locate each notation feature by position.
- State the local meaning of each feature.
- Check the complete species notation.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 25 — Separate observation from inference

- Record the observation literally.
- State the claim separately.
- Explain exactly how the observation supports the claim.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 26 — Separate observation from inference

- Record the observation literally.
- State the claim separately.
- Explain exactly how the observation supports the claim.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 27 — Separate observation from inference

- Record the observation literally.
- State the claim separately.
- Explain exactly how the observation supports the claim.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 28 — Track the same reacting species

- Identify reactant species.
- Track each corresponding species after the process.
- Mark changed and unchanged species.

- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 29 — Track the same reacting species

- Identify reactant species.
- Track each corresponding species after the process.
- Mark changed and unchanged species.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 30 — Track the same reacting species

- Identify reactant species.
- Track each corresponding species after the process.
- Mark changed and unchanged species.
- Check that the overall charge is preserved or interpreted correctly.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 31 — Translate between particles and symbols

- Identify the starting representation level.
- Mark invariant entities/counts.
- Translate one feature at a time.
- Cross-check against the starting representation.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 32 — Translate between particles and symbols

- Identify the starting representation level.
- Mark invariant entities/counts.
- Translate one feature at a time.
- Cross-check against the starting representation.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 33 — Translate between particles and symbols

- Identify the starting representation level.

- Mark invariant entities/counts.
- Translate one feature at a time.
- Cross-check against the starting representation.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 34 — Verify a chemical representation before accepting it

- Name the required verification checks.
- Run each check independently.
- Repair any failed check and re-run it.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 35 — Verify a chemical representation before accepting it

- Name the required verification checks.
- Run each check independently.
- Repair any failed check and re-run it.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Solution 36 — Verify a chemical representation before accepting it

- Name the required verification checks.
- Run each check independently.
- Repair any failed check and re-run it.
- Check that the same chemical species is being tracked.
- Check that the final conclusion follows from the evidence and rule used.

A complete response states the relevant chemical evidence or rule, executes the recorded reasoning route, and gives the conclusion only after the required checks pass.

No additional condition or exception is required for this task.

Appendix C — Printable Handout

Answer-free printable reference. It introduces no new capability.

Attach a role only after tracking the species

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that the overall charge is preserved or interpreted correctly.

Check atom conservation explicitly

First move: Identify the target.

Rule / decision cue: Choose the conservation check required by the question.

Verification cue: Count each required atom on both sides.

Check the rule, condition and exception

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that every stated condition or exception is still attached.

Classify a change from the stated evidence

First move: Identify the target.

Rule / decision cue: Classify only from the stated evidence and the rule taught here.

Verification cue: Check that the final conclusion follows from the evidence and rule used.

Separate ionic charge from subscripts

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that the overall charge is preserved or interpreted correctly.

Keep reaction conditions attached

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that every stated condition or exception is still attached.

Connect apparatus, property and method

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that the final conclusion follows from the evidence and rule used.

Read a chemical formula before calculating

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that the overall charge is preserved or interpreted correctly.

Separate observation from inference

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that the final conclusion follows from the evidence and rule used.

Track the same reacting species

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that the overall charge is preserved or interpreted correctly.

Translate between particles and symbols

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that the same chemical species is being tracked.

Verify a chemical representation before accepting it

First move: Identify the target.

Rule / decision cue: Use only the chemistry stated or taught for this task.

Verification cue: Check that the same chemical species is being tracked.